

---

# **Dealing with Limited Overlap in Estimation of Average Treatment Effects**

Replication and Extensions of Crump et al. (2009)

Hayley Shin, Carolyn Ha, Julianna Pellicano

---

# Introduction: Limited Overlap and Treatment Effect Estimation

**Paper:** Dealing with Limited Overlap in Estimation of Average Treatment Effects (Crump, Hotz, Imbens & Mitnik (2009))

**Core Problem:** How should we handle poor treatment/control overlap? The below equation represents the propensity score, i.e. the probability of receiving treatment conditional on covariates  $X$ . When  $e(X)$  is near 0 or 1, comparable treated/control units are limited, making treatment effect estimates unstable.

$$e(X) = P(W = 1 \mid X)$$

## Paper Contributions:

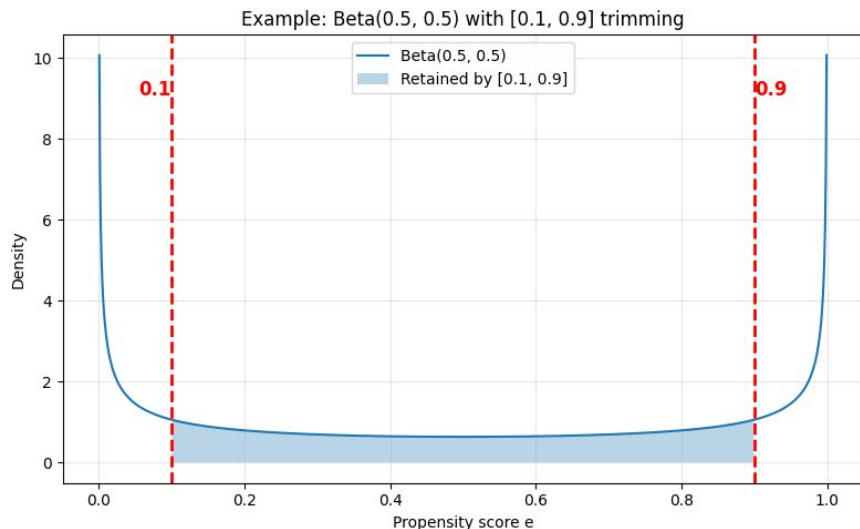
- They begin with a general weighting function, and then restrict to 0/1 weights, turning the problem into choosing the subset with the best overlap and lowest asymptotic variance.
- Shows that, under homoskedasticity, the optimal rule has the form:  $\alpha \leq e(X) \leq 1 - \alpha$
- Motivates the common rule of thumb:  $0.1 \leq e(X) \leq 0.9$

## Our Project:

- Replicate the paper's simulation and empirical results
- Extend the framework by exploring the case of heterogeneous outcome variance as well as by comparing hard trimming with smoother weighting rules

# Repl. 1

## How Good Is Simple Propensity-Score Trimming?



Beta(0.5, 0.5): many extreme propensity scores  $\rightarrow$  poor overlap  
Shaded region = observations retained by [0.1, 0.9]

- **Goal:** Does a simple fixed trimming rule work well across different propensity-score distributions?
- **Setup:**  $e(X) \sim \text{Beta}(\beta, \gamma)$   
Different  $(\beta, \gamma) \rightarrow$  different overlap patterns
- **Trimming:**  
$$\alpha \leq e(X) \leq 1 - \alpha$$
- **Benchmark:**  
$$\alpha^* = \arg \min_{\alpha} V_{S, \alpha}$$

Compare fixed cutoffs (.01, .10) with the optimal symmetric cutoff  $\alpha^*$

# Repl. 1

## The 0.1 Rule Is Nearly Optimal

Our replication of table 1

Original Table 1: Variance ratios for beta distributions

| $\beta$ |                                                             | $\gamma$ |       |       |       |
|---------|-------------------------------------------------------------|----------|-------|-------|-------|
|         |                                                             | 0.5      | 1.0   | 2.0   | 4.0   |
| 0.5     | $V_S(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$        | 13.38    | 11.68 | 13.71 | 12.83 |
|         | $V_{S,0.01}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ | 1.70     | 1.64  | 1.71  | 1.58  |
|         | $V_{S,0.10}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ | 1.00     | 1.00  | 1.00  | 1.04  |
| 1.0     | $V_S(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$        |          | 2.68  | 2.65  | 3.36  |
|         | $V_{S,0.01}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ |          | 1.39  | 1.39  | 1.47  |
|         | $V_{S,0.10}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ |          | 1.00  | 1.00  | 1.01  |
| 2.0     | $V_S(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$        |          |       | 1.11  | 1.16  |
|         | $V_{S,0.01}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ |          |       | 1.09  | 1.12  |
|         | $V_{S,0.10}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ |          |       | 1.00  | 1.00  |
| 4.0     | $V_S(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$        |          |       |       | 1.02  |
|         | $V_{S,0.01}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ |          |       |       | 1.02  |
|         | $V_{S,0.10}(\gamma, \beta) / V_{S,\alpha^*}(\gamma, \beta)$ |          |       |       | 1.00  |

Our replication

| $\beta$ | Metric               | $\gamma = 0.5$ | $\gamma = 1.0$ | $\gamma = 2.0$ | $\gamma = 4.0$ |
|---------|----------------------|----------------|----------------|----------------|----------------|
| 0.5     | $\alpha^*$           | 0.0939         | 0.0959         | 0.0897         | 0.0680         |
|         | $V_{0.10} / V_{opt}$ | 1.0008         | 1.0003         | 1.0024         | 1.0374         |
| 1       | $\alpha^*$           |                | 0.1018         | 0.1018         | 0.0843         |
|         | $V_{0.10} / V_{opt}$ |                | 1.0001         | 1.0001         | 1.0063         |
| 2       | $\alpha^*$           |                |                | 0.1127         | 0.1073         |
|         | $V_{0.10} / V_{opt}$ |                |                | 1.0013         | 1.0006         |
| 4       | $\alpha^*$           |                |                |                | 0.1249         |
|         | $V_{0.10} / V_{opt}$ |                |                |                | 1.0012         |

- Across all 10 Beta settings

$$1.0001 \lesssim \frac{V_{0.10}}{V_{\alpha^*}} \leq 1.0374$$

**Takeaway:** Under homoskedasticity, [0.1,0.9] is an excellent practical approximation to the optimal symmetric trimming rule.



But why does this simple rule work so well?

## Ext. 1

### What If Outcome Variance is Heterogeneous?

Original: Homoscedasticity

$$\sigma_1^2(X) = \sigma_0^2(X) = \sigma^2$$
$$k(X) = \frac{\sigma^2}{e(X)} + \frac{\sigma^2}{1 - e(X)} = \frac{\sigma^2}{e(X)(1 - e(X))}$$

- Variance contribution depends **only on the propensity score**
  - $k(e)$  is symmetric around  $e=0.5$
- ⇒ optimal retained set:  $[\alpha, 1 - \alpha]$

Extension: Heteroskedasticity

$$k(X) = \frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1 - e(X)}$$

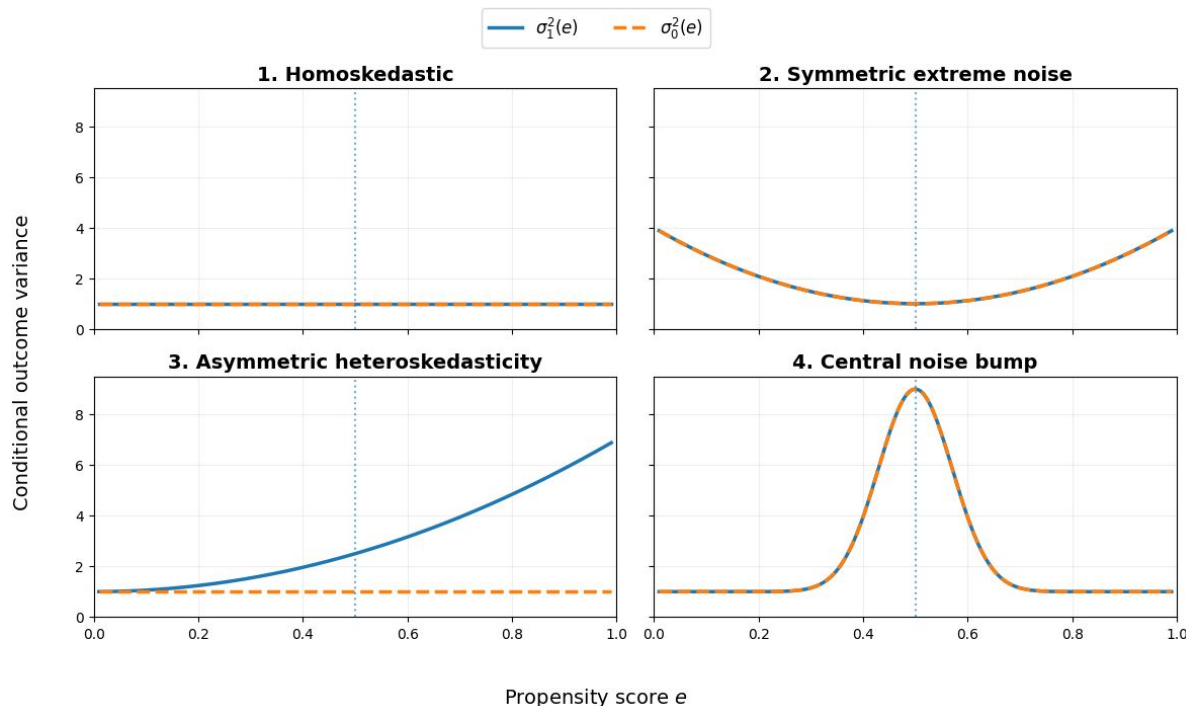
- Variance contribution depends on both **overlap + outcome variability**
- Propensity score alone may no longer identify the lowest-variance observations
- **Good overlap**  $\nRightarrow$  **low variance contribution**

Is propensity-score-only trimming still efficient when conditional outcome variance varies with X?

# Ext. 1

## Simulation Design:

### How Do We Introduce Heteroskedasticity



- Hold the propensity-score distribution fixed, and vary only the conditional outcome variance structure
- For this simulation, we model  $\sigma_w^2(X)$  as a function of  $e(X)$

# Ext. 1

## Results - Robustness Across All 10 Beta Settings

Range of relative variance across the 10 Beta distributions

| Scenario                      | Fixed<br>0.10 Min | Fixed<br>0.10<br>Max | Best<br>symmetric<br>Min | Best<br>symmetric<br>Max |
|-------------------------------|-------------------|----------------------|--------------------------|--------------------------|
| Homoskedastic                 | 1.000             | 1.037                | 1.000                    | 1.000                    |
| Symmetric extreme noise       | 1.041             | 1.154                | 1.000                    | 1.000                    |
| Asymmetric heteroskedasticity | 1.001             | 1.109                | 1.000                    | 1.006                    |
| Central noise bump            | 1.210             | 1.312                | 1.046                    | 1.234                    |

Each entry reports the min/max of  $V_{\text{rule}}/V_{k\text{-opt}}$  across the 10 Beta settings.  
(1.00 = variance-aware optimum)

- **Symmetric extreme noise:** Re-optimizing the cutoff is enough.
- **Asymmetric heteroskedasticity:** The variance-aware optimum is no longer symmetric, but a symmetric rule often remains a good approximation.
- **Central noise bump:** Even the best symmetric interval can be inefficient.

The strongest departure occurs when high outcome variance appears in the region of good overlap.

## Ext. 1

### Results - Illustrative case: $e(X) \sim \text{Beta}(0.5, 0.5)$

How does the optimal retained set change?

| Scenario                      | Best symmetric $\alpha^*$ | Fixed [0.1,0.9] Rel. variance | Best symmetric Rel. variance | Variance-aware retained set          | Retained fraction (k-opt) |
|-------------------------------|---------------------------|-------------------------------|------------------------------|--------------------------------------|---------------------------|
| Homoskedastic                 | 0.0939                    | 1.0008                        | 1.0000                       | [0.094, 0.906]                       | 0.603                     |
| Symmetric extreme noise       | 0.1936                    | 1.1495                        | 1.0000                       | [0.194, 0.806]                       | 0.420                     |
| Asymmetric heteroskedasticity | 0.0655                    | 1.0340                        | 1.0060                       | [0.057, 0.920]                       | 0.664                     |
| Central noise bump            | 0.0399                    | <b>1.3109</b>                 | <b>1.1257</b>                | [0.061, 0.404] $\cup$ [0.596, 0.939] | 0.560                     |

*Relative variance is measured against the variance-aware optimum k-opt; 1.00 = optimal.*

Heteroskedasticity can shift the cutoff → break symmetry → make a single central interval suboptimal

## Ext. 2

### Original Hard-Trimming Setup

$$\tau_{S,\omega} = \frac{\sum_{i=1}^N \tau(X_i) \omega(X_i)}{\sum_{i=1}^N \omega(X_i)},$$

Weighted Treatment Effect



$$\omega(x) = 1\{x \in A\}$$

Restricting to Indicator weights turns the problem into choosing a subset  $A^*$



$$\omega(e) = 1\{\alpha \leq e \leq 1 - \alpha\}.$$

Resulting Weight Function  
(hard-trimming rule)

$$V_{\omega}^{\text{eff}} = \frac{1}{E[\{\omega(X)\}^2]} E \left[ \omega(X)^2 \left\{ \frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1 - e(X)} \right\} \right].$$

General Efficiency Bound

$$\mathbb{A}_H^* = \{x \in \mathbb{X} \mid \alpha \leq e(x) \leq 1 - \alpha\}$$

Under homoskedasticity, the optimal subset is a propensity score interval

General Weighting  
Framework

Restriction to  
Hard-Trimming

## Ext. 2

## Softening Cutoffs with Smooth Weighting and Boundary Smoothing

$$V_{\omega} \propto \frac{E[\omega(e)^2 (\frac{1}{e} + \frac{1}{1-e})]}{E[\omega(e)]^2}$$

Efficiency Bound under  
homoskedasticity



$$\omega(e) \propto e(1-e)$$



$$\omega(e) = e(1-e)$$

Variance-minimizing smooth weight  
function without indicator restriction

### Smooth Weighting

Smooth transition near 0.1 and 0.9 with width delta

$$\omega_{\delta}(e) = S\left(\frac{e - (0.1 - \delta)}{2\delta}\right) S\left(\frac{(0.9 + \delta) - e}{2\delta}\right)$$

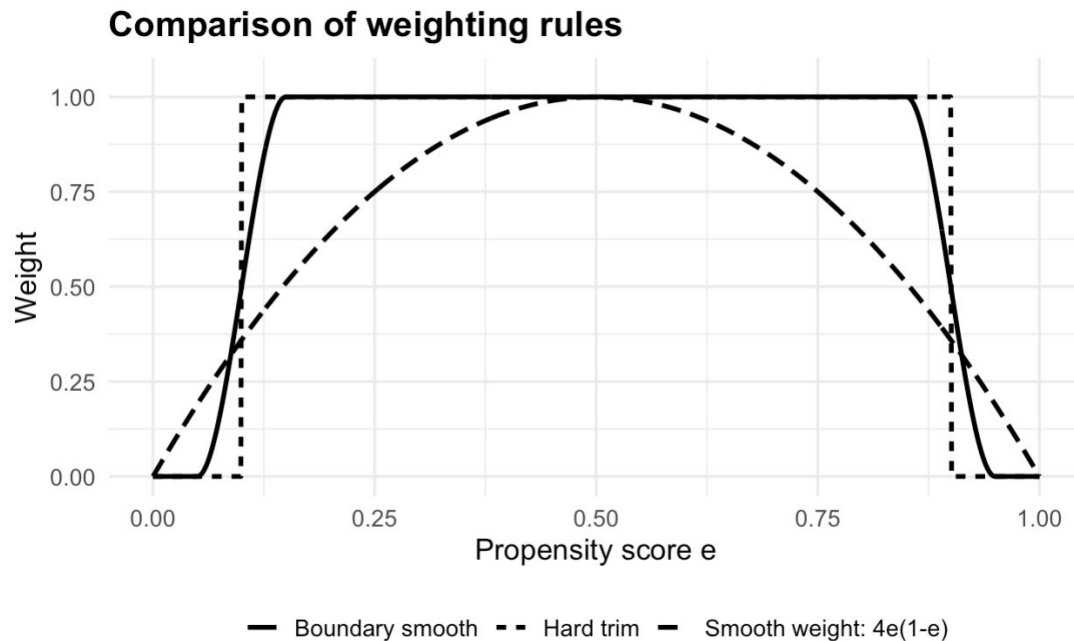


$$\omega_{\delta}(e) \rightarrow 1\{0.1 \leq e \leq 0.9\} \quad \text{as } \delta \rightarrow 0$$

### Boundary Smoothing

Inspired by Yang & Ding 2018

## Ext. 2



## Ext. 2

## Results: Smooth weighting helps most under limited overlap

We simulated the same Beta propensity score distributions by Crump et al. (Section 3.3) and maintain the assumption of homoscedasticity

|   | Scenario                      | Method                         | Relative Efficiency Bound | ESS/N | TVD  |
|---|-------------------------------|--------------------------------|---------------------------|-------|------|
| 1 | Beta(0.5, 4) – Poor overlap   | Hard 0.10 trim                 | 1.04                      | 0.37  | 0.63 |
| 2 | Beta(0.5, 4) – Poor overlap   | Boundary smooth (delta = 0.05) | 0.90                      | 0.42  | 0.56 |
| 3 | Beta(0.5, 4) – Poor overlap   | Smooth weight e(1-e)           | 0.73                      | 0.53  | 0.40 |
| 4 | Beta(1, 1) – Moderate overlap | Hard 0.10 trim                 | 1.00                      | 0.80  | 0.20 |
| 5 | Beta(1, 1) – Moderate overlap | Boundary smooth (delta = 0.05) | 0.94                      | 0.83  | 0.18 |
| 6 | Beta(1, 1) – Moderate overlap | Smooth weight e(1-e)           | 0.87                      | 0.83  | 0.19 |
| 7 | Beta(4, 4) – Strong overlap   | Hard 0.10 trim                 | 1.00                      | 0.99  | 0.01 |
| 8 | Beta(4, 4) – Strong overlap   | Boundary smooth (delta = 0.05) | 1.00                      | 1.00  | 0.01 |
| 9 | Beta(4, 4) – Strong overlap   | Smooth weight e(1-e)           | 0.97                      | 0.98  | 0.06 |

**Relative efficiency bound:** normalized by the optimal hard trim; lower is better.

**ESS/N:** effective sample size ratio; higher is better.

**TVD:** change in the propensity-score distribution; lower is less target-population shift.

$$\frac{ESS}{N} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \cdot \frac{1}{N}$$

### Key Takeaways:

1. **Gains are largest when overlap is poor** (smooth weighting reduces efficiency bound while retaining more effective information and less propensity-score distribution shifting)
2. **Boundary smoothing is an intermediate option** (smaller efficiency gains but still offers improvement, especially when overlap is poor)
3. **Little to no gain when overlap is strong** (similar efficiency with smoothing but more propensity score distribution shift)

# Repl. 2

## Empirical replication

Following the empirical reanalysis in Crump et al. (2009)

|                  |                                                                                                                                                                                                                                                                    |                         |                                                                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Data</b>      | Observational Data on <b>Right Heart Catheterization (RHC)</b><br>( <i>RHC is a diagnostic procedure used for critically ill patients</i> )                                                                                                                        |                         |                                                                                                           |
| <b>Goal</b>      | Estimate the effect of receiving RHC on 30 day survival                                                                                                                                                                                                            |                         |                                                                                                           |
| <b>Variables</b> | $W_i = \begin{cases} 1 & \text{if patient } i \text{ received RHC} \\ 0 & \text{otherwise} \end{cases}$ $Y_i = \begin{cases} 1 & \text{if patient } i \text{ survived at 30 days} \\ 0 & \text{otherwise} \end{cases}$ $X_i = 72 \text{ pre-treatment covariates}$ | <b>Sample</b>           | N = 5,735<br>Treated = 2,184<br>Controls = 3,551                                                          |
|                  |                                                                                                                                                                                                                                                                    | <b>Propensity Score</b> | Estimated using logistic regression with all 72 covariates:<br>$\hat{e}(X_i) = \hat{P}(W_i = 1 \mid X_i)$ |

# Repl. 2

## Empirical replication

### Data Construction

#### Data Issue

Initial Attempt → used the R package ATbounds RHC data

BUT our estimates differed substantially from the paper → Cross-check with raw data

| Cross-check of ATbounds outcome variable | Mismatched Obs. # | Result         |
|------------------------------------------|-------------------|----------------|
| Raw <i>30 day survival</i> variable      | 1,804 / 5,735     | Does not match |
| Raw <i>death</i> variable                | 0 / 5,735         | Exact match    |

Treatment variable: 0 mismatches between raw data and ATbounds R package

► possible data processing error

#### Data Sources

R package ATbounds RHC data: Treatment + 72 covariates

Raw Vanderbilt RHC data: Outcome (30 day survival variable)

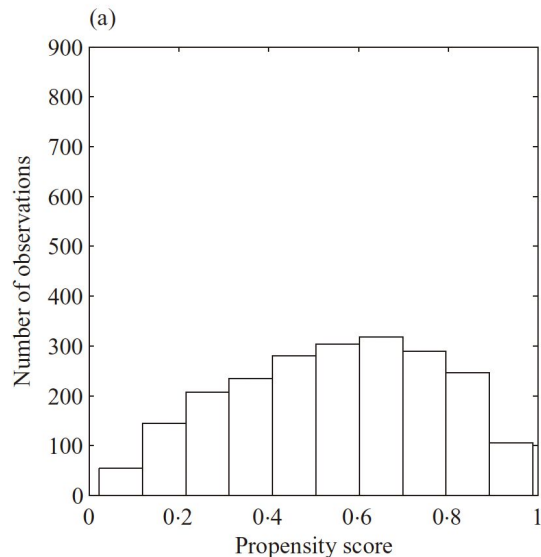
# Repl. 2

## Empirical replication

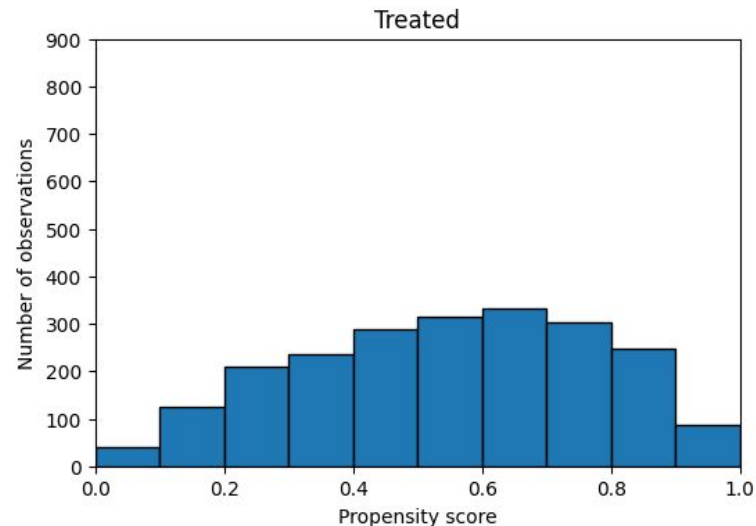
Figure 1 replication: estimated propensity score distributions

### Treated Histogram

#### Original Figure



#### Replication Figure



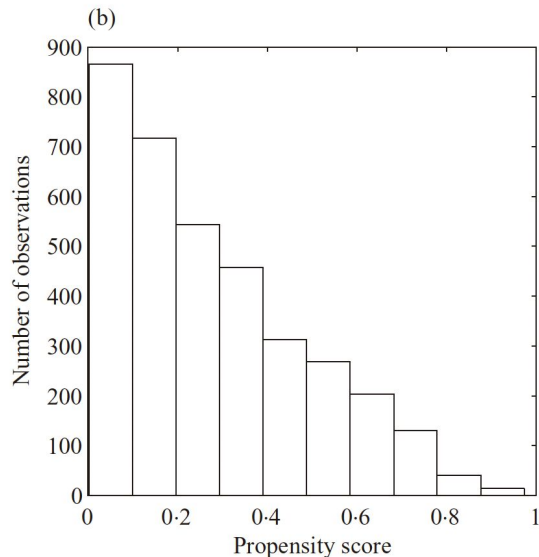
# Repl. 2

## Empirical replication

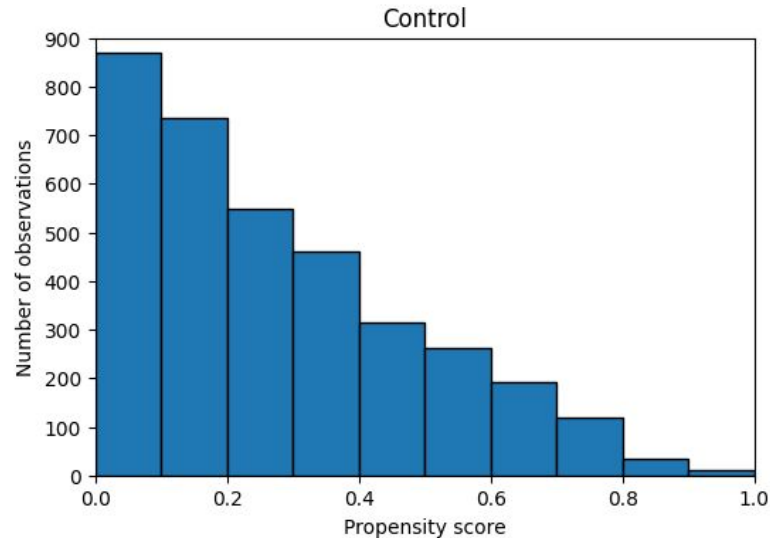
Figure 1 replication: estimated propensity score distributions

### Control Histogram

#### Original Figure



#### Replication Figure



# Repl. 2

## Empirical replication

### Sample construction and Table 2 replication

#### Sample Definitions

**Full Sample:**  $N = 5,735$

**Rule of thumb trim:**  
 $0.1 \leq \hat{e}(X_i) \leq 0.9$

**Optimal trim:**  
 $\hat{\alpha} \leq \hat{e}(X_i) \leq 1 - \hat{\alpha}$

#### Optimal cut-off

Chosen to minimize the asymptotic variance of the trimmed-sample ATE

**Paper:**  $\alpha = 0.1026$

**Replication:**  $\hat{\alpha} = 0.1029$

#### Original Table

*Table 2. Subsample sizes for right heart catheterization data*

|          | $\hat{e}(X_i) < 0.1$ | $0.1 \leq \hat{e}(X_i) \leq 0.9$ | $0.9 < \hat{e}(X_i)$ | All  |
|----------|----------------------|----------------------------------|----------------------|------|
| Controls | 870                  | 2671                             | 10                   | 3551 |
| Treated  | 40                   | 2057                             | 87                   | 2184 |
| All      | 910                  | 4728                             | 97                   | 5735 |

#### Replication Table

*Table 2. Subsample sizes for right heart catheterization data*

|          | $\hat{e}(X_i) < 0.1$ | $0.1 \leq \hat{e}(X_i) \leq 0.9$ | $0.9 < \hat{e}(X_i)$ | All  |
|----------|----------------------|----------------------------------|----------------------|------|
| Controls | 870                  | 2671                             | 10                   | 3551 |
| Treated  | 40                   | 2057                             | 87                   | 2184 |
| All      | 910                  | 4728                             | 98                   | 5735 |

# Repl. 2

## Empirical replication

### Table 3 replication

#### Original Table

Table 3. Estimates for average treatment effects in right heart catheterization study

|                                     | Estimate | SE(1)  | SE(2)  |
|-------------------------------------|----------|--------|--------|
| Full sample                         | −0.0593  | 0.0166 | 0.0167 |
| $\hat{e}(X_i) \in [0.1, 0.9]$       | −0.0590  | 0.0143 | 0.0143 |
| $\hat{e}(X_i) \in [0.1026, 0.8974]$ | −0.0601  | 0.0143 | 0.0144 |

#### Replication Table

|                                     | Estimate | SE(1)  | SE(2)  |
|-------------------------------------|----------|--------|--------|
| Full sample                         | −0.0593  | 0.0166 | 0.0165 |
| $\hat{e}(X_i) \in [0.1, 0.9]$       | −0.0590  | 0.0143 | 0.0143 |
| $\hat{e}(X_i) \in [0.1026, 0.8974]$ | −0.0599  | 0.0143 | 0.0143 |
| $\hat{e}(X_i) \in [0.1029, 0.8971]$ | −0.0599  | 0.0144 | 0.0144 |

#### Estimation within each sample

- Re-estimate propensity scores using all 72 covariates
- Estimate the normalized Horvitz-Thompson ATE
- Bootstrap SE(1) and SE(2) using 50,000 replications

#### Normalized Horvitz–Thompson estimator:

$$\hat{\tau} = \frac{\sum_{i=1}^N \frac{W_i Y_i}{\hat{e}(X_i)}}{\sum_{i=1}^N \frac{W_i}{\hat{e}(X_i)}} - \frac{\sum_{i=1}^N \frac{(1 - W_i) Y_i}{1 - \hat{e}(X_i)}}{\sum_{i=1}^N \frac{1 - W_i}{1 - \hat{e}(X_i)}}$$

# Repl. 2

## Empirical replication

### Table 3 replication

#### Original Table

Table 3. Estimates for average treatment effects in right heart catheterization study

|                                     | Estimate | SE(1)  | SE(2)  |
|-------------------------------------|----------|--------|--------|
| Full sample                         | −0.0593  | 0.0166 | 0.0167 |
| $\hat{e}(X_i) \in [0.1, 0.9]$       | −0.0590  | 0.0143 | 0.0143 |
| $\hat{e}(X_i) \in [0.1026, 0.8974]$ | −0.0601  | 0.0143 | 0.0144 |

#### Replication Table

|                                     | Estimate | SE(1)  | SE(2)  |
|-------------------------------------|----------|--------|--------|
| Full sample                         | −0.0593  | 0.0166 | 0.0165 |
| $\hat{e}(X_i) \in [0.1, 0.9]$       | −0.0590  | 0.0143 | 0.0143 |
| $\hat{e}(X_i) \in [0.1026, 0.8974]$ | −0.0599  | 0.0143 | 0.0143 |
| $\hat{e}(X_i) \in [0.1029, 0.8971]$ | −0.0599  | 0.0144 | 0.0144 |

### Estimation within each sample

- Re-estimate propensity scores using all 72 covariates
- Estimate the normalized Horvitz-Thompson ATE
- Bootstrap SE(1) and SE(2) using 50,000 replications

### Bootstrap SE (B = 50,000):

$$SE(1) = SD(\hat{t}^{*(1)}, \dots, \hat{t}^{*(B)})$$

$$SE(2) = \frac{\hat{t}_{0.95}^* - \hat{t}_{0.05}^*}{2 \times 1.645}$$

## Repl. 2 + Ext. 2

### Empirical extension

#### Boundary Smoothing and Smooth Weighting

|                                    | Estimate | SE(1)         | SE(2)         | ESS / N       | TVD           |
|------------------------------------|----------|---------------|---------------|---------------|---------------|
| Rule of thumb [0.1, 0.9]           | -0.0590  | 0.0143        | 0.0143        | 0.8244        | 0.1756        |
| Boundary smoothing, $\delta = .01$ | -0.0596  | 0.0147        | 0.0147        | 0.8291        | 0.1716        |
| Boundary smoothing, $\delta = .05$ | -0.0645  | 0.0141        | 0.0141        | 0.8522        | <b>0.1536</b> |
| Smooth weighting $e(1 - e)$        | -0.0654  | <b>0.0134</b> | <b>0.0133</b> | <b>0.8580</b> | 0.1728        |

**ESS/N:** effective sample size ratio; higher is better (many observations contribute with relatively balanced weights)

**TVD:** total variation distances; change in the propensity-score distribution; lower is less target-population shift

### Key Takeaways:

1. Smooth weighting achieves the greatest precision gain and highest ESS/N
2. Boundary Smoothing ( $\delta = .05$ ) provides the smallest distributional shift while also slightly improving precision

---

**Thank you**

---